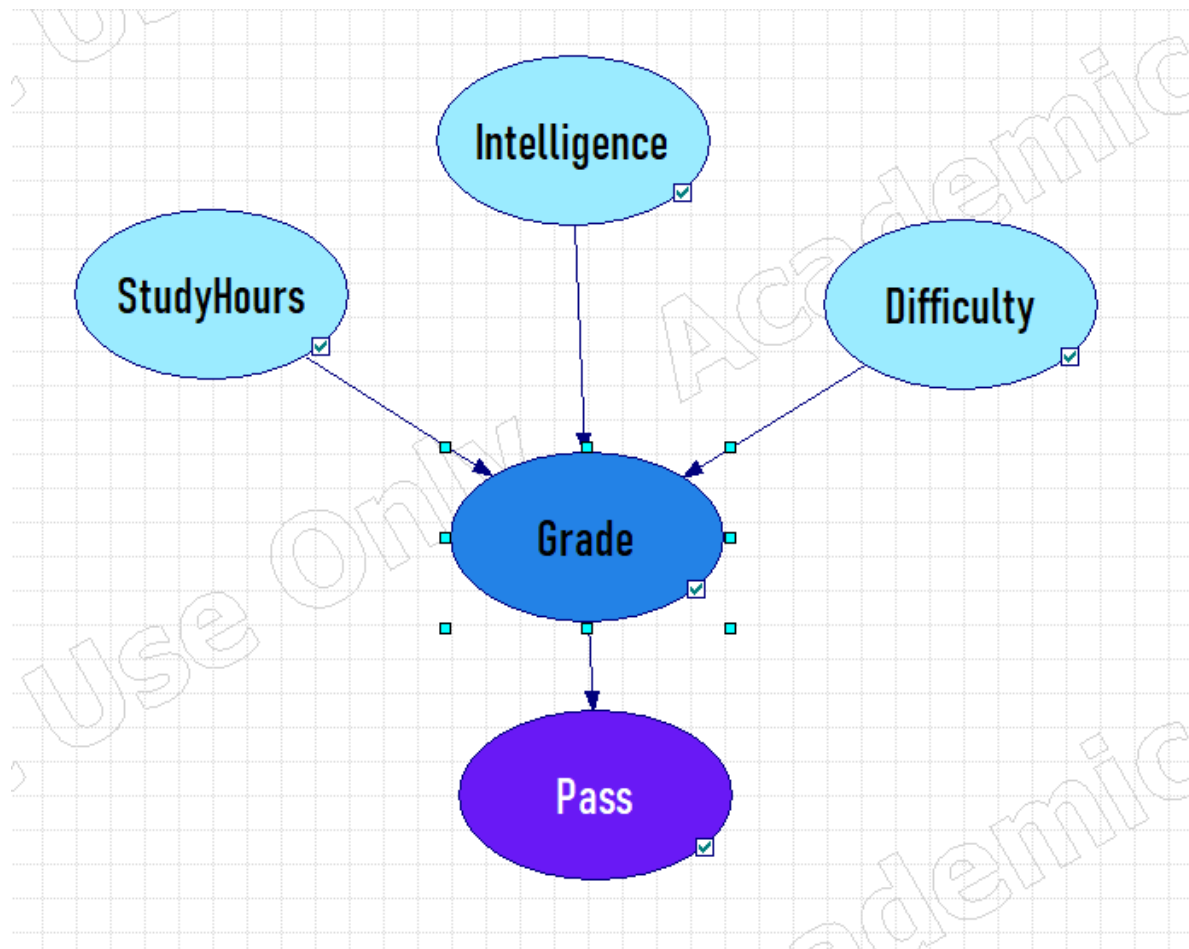


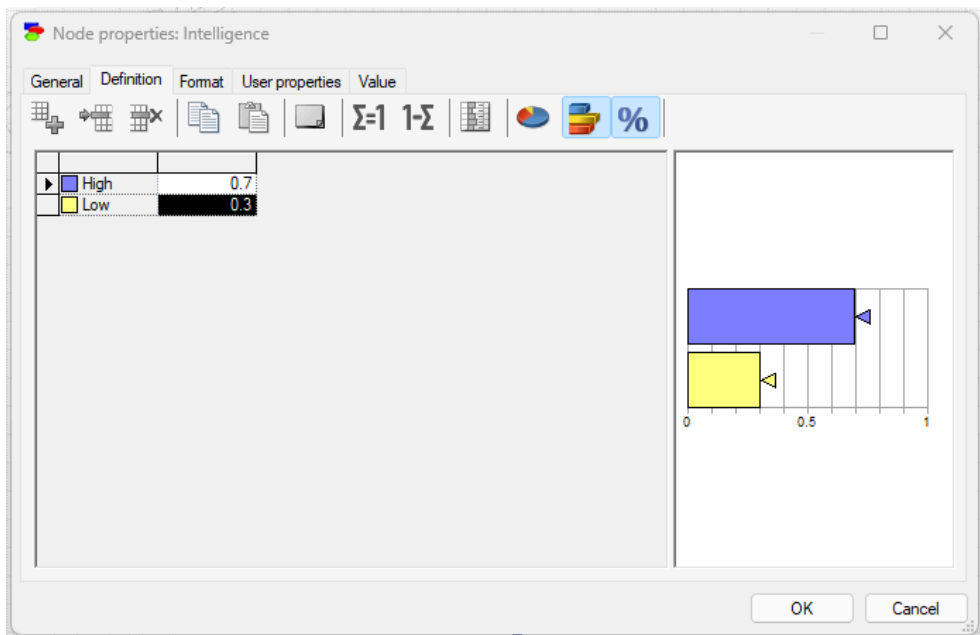
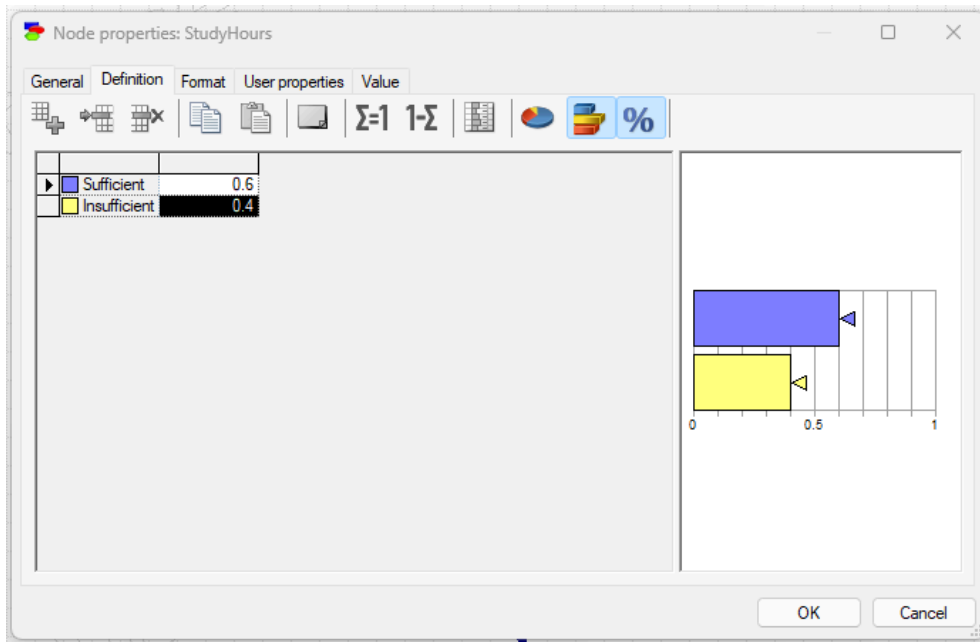
24K-1037

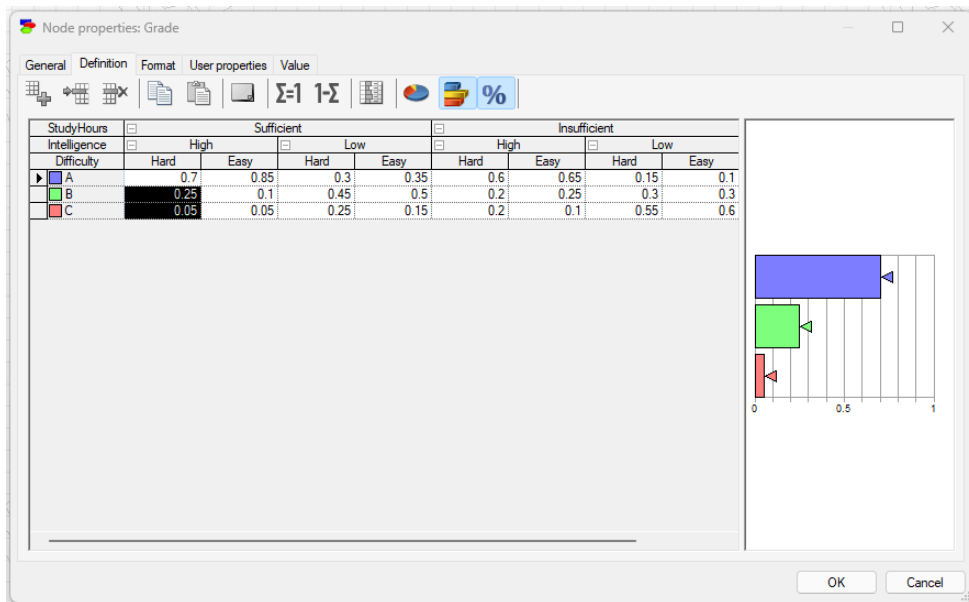
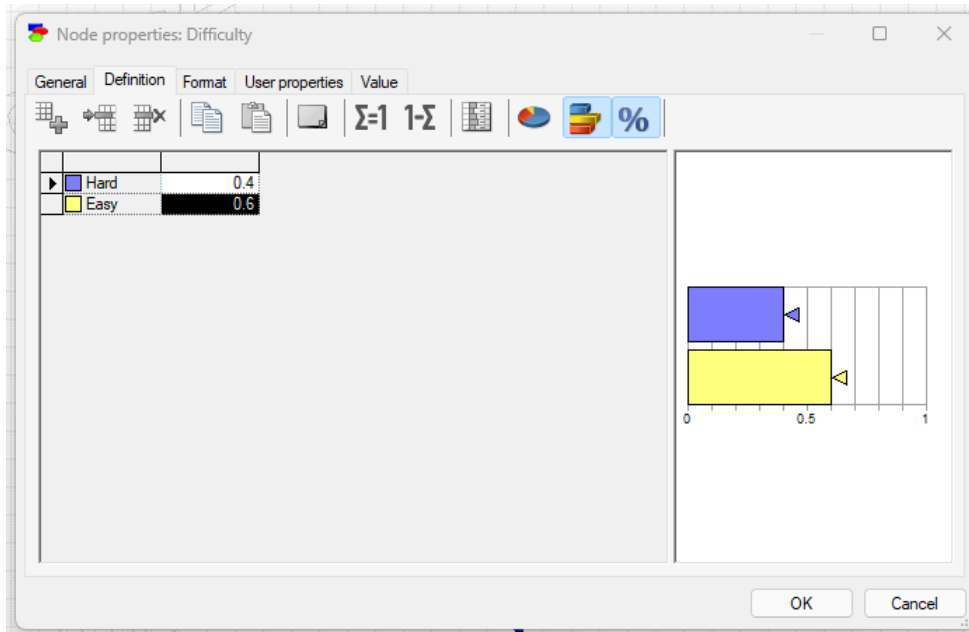
TASK 1

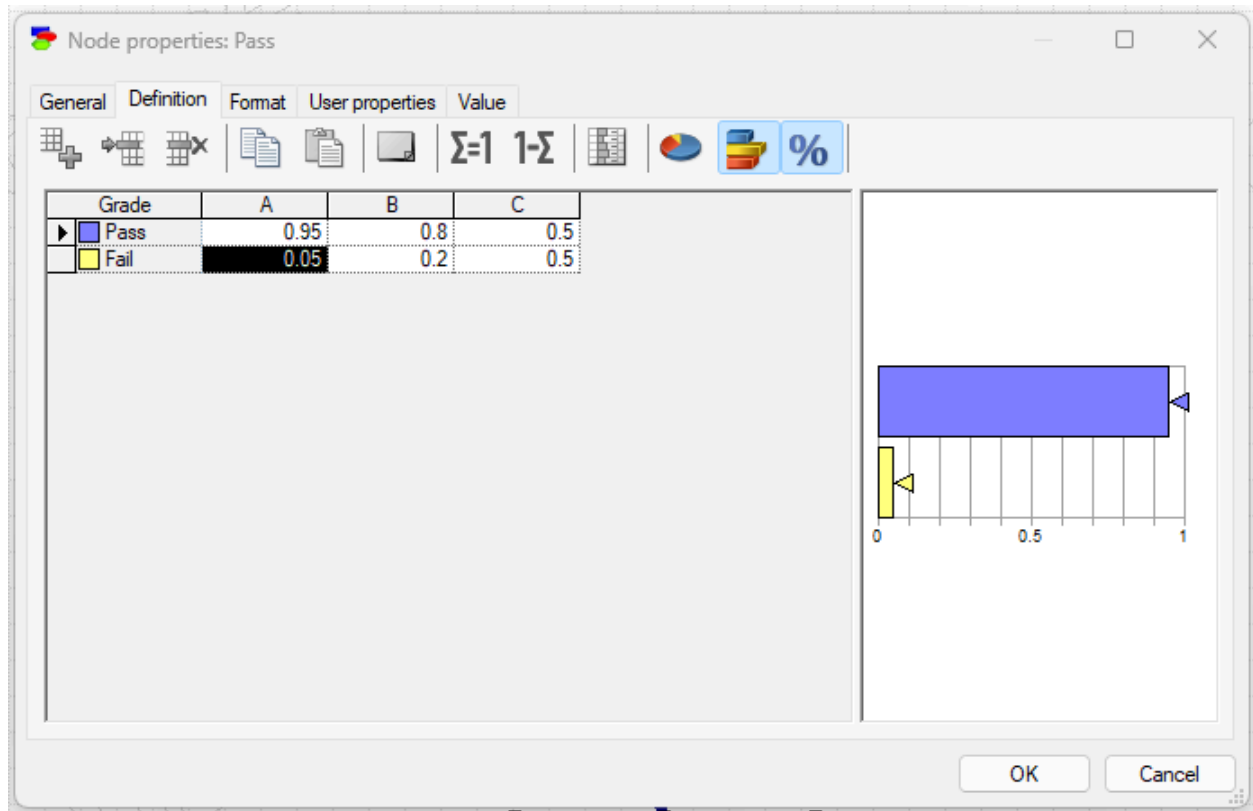
1. $P(\text{Red})$: $26/52 = \frac{1}{2}$
2. Reds: 26
Red Hearts: 13
 $P(\text{Heart}|\text{Red})$: $13/26 = \frac{1}{2}$
3. Face Cards: 12
Diamond Faces: 3
 $P(\text{Diamond}|\text{Face})$: $3/12 = \frac{1}{4}$
4. Face Cards: 12
Faced Spades & Queens (total): $3 + 4 - 1 = 6$
 $P(\text{Spade} \vee \text{Queen}|\text{Face})$: $6/12 = \frac{1}{2}$

TASK 2









Pass	phi(Pass)
Pass(0)	0.8540
Pass(1)	0.1460

TASK 4

Sequence: Sunny -> Rainy -> Cloudy -> Cloudy -> Rainy -> Rainy -> Cloudy -> Cloudy -> Cloudy -> Cloudy
 Prob of >= 3 rainy days: 0.3809

- Ipython file included for tasks 2 & 4